An Improved Scheme for High-Resolution Point Cloud Map Generation Based on FMCW Radar

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***Abstract*—For the frequency modulated continuous wave (FMCW) radar, the angle resolution of the generated point cloud map (PCM) is limited by both the number of receiver antennas and the interference of multi-target environments. This paper proposes an improved scheme for high-resolution PCM generation based on a 77 GHz FMCW radar system. The scheme consists of a method to synthesize an equivalent virtual array based on the time-division multiplexing (TDM) multiple-input multiple-output (MIMO) technique and an improved target detection algorithm based on multi-channel fusion. The results show that the proposed scheme could suppress the noise floor and improve the accuracy of target detection by 1.33 times compared to a conventional constant false alarm rate (CFAR) detector in a multi-target environment. Moreover, by applying the proposed scheme, the angle resolution of the PCM generated by the radar system reaches to 1.3022 degrees.**

***Keywords*—FMCW radar, TDM-MIMO, virtual array, high-resolution, point cloud map (PCM)**

I. Introduction

In recent years, frequency modulated continuous wave (FMCW) radar has attracted considerable attention due to its superior performance in target detection and localization [1], and high reliability in harsh environmental conditions. For FMCW radars, fast-ramp based two-dimensional (2D) fast Fourier transform (FFT) is an effective and established method to estimate the range and velocity information of various targets [2]. However, in multi-target and highly cluttered environments, the FMCW radar faces ambiguity problems related to the separation of targets [4]. Although a weak moving target can be detected in unwanted strong clutter using a coherent phase difference method [5], this method is not useful for weak stationary target detection. Multi-target situations also lead to a masking effect, which has a negative influence on determining the number of actual targets and can result in the detection of ghost targets [6].

In addition, the angle of arrive (AOA) of targets can be estimated using discrete Fourier transform (DFT) based on the phase difference of echo signals among receiver antennas [7]. Thus, the angle resolution is limited by the number of receiver antennas. Although a novel AOA estimation framework without restrictions on the number of antennas has been proposed [9], its computational load is too high to perform in real time. In view of these limitations, this paper proposes an improved scheme for high-resolution point cloud map (PCM) generation. First, the constant false alarm rate (CFAR) detector based on multi-channel fusion is proposed, which performs better in multi-target situations than conventional CFAR detectors. Meanwhile, the number of virtual receiver antennas are enhanced dramatically by synthesizing a virtual array based on the time-division multiplexing (TDM) multiple-input multiple-output (MIMO) technique [10], to improve the angle resolution. Finally, a high-resolution PCM is generated for specific scenario using the proposed scheme.

This paper is organized as follows. The basic model of an FMCW radar signal and the method to synthesize a virtual antenna array are described in Section II. In Section III, the proposed scheme is theoretically analyzed. The framework of the developed radar system is then briefly introduced in Section IV. In Section V, the developed radar system is used to verify the effectiveness of the proposed scheme. Finally, conclusions are presented in Section VI.

II. FMCW Radar Principle Based on TDM-MIMO

A. FMCW Radar Signal Model

As seen in Fig. 1, the transmitted FMCW chirp signal is a wideband non-stationary signal [9], whose frequency is proportional to the time characteristic, and can be modeled as:

$$x_{tx}(t) = Acos(2\pi(f_c t + St^2/2) + \varphi_0) \text{ for } 0 \le t \le T, \quad (1)$$

where A, f_c, S, φ_0, and T are the amplitude, the starting frequency, the frequency modulated slope, the initial phase, and the time duration of the transmitted signal, respectively.

The echo signal reflected from a specific target will be received by the receiver antenna and can be expressed as:

$$x_{rx}(t) = A_r cos(2\pi(f_c(t-\tau) + S(t-\tau)^2/2) + \varphi_0), \quad (2)$$

where A_r and τ represent the echo signal amplitude and the time of flight (TOF).

B. Synthesis of the Equivalent Virtual Antenna Array

In this radar system, the angle resolution is denoted as:

$$\Delta\theta = \lambda / N_{rx} d cos\theta, \quad (3)$$

where λ, N_{rx}, and d are the wavelength, the number of receiver antennas, and the interval between receiver antennas.

To improve the angle resolution, the number of receiver antennas is increased. The antenna array architecture of the

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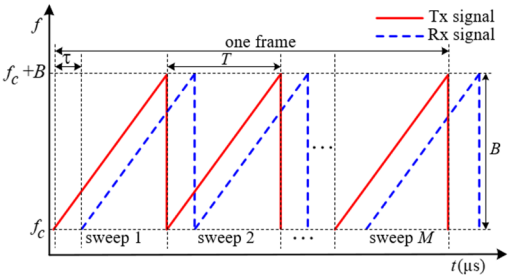

Fig. 1. The illustration of the transmitted signal and received signal.

radar system is then designed, as illustrated in Fig.2. The designed MIMO antenna array is a uniform linear array (ULA), which consists of 9 transmitter (TXs) antennas and 16 receiver antennas (RXs).

The foundational principle of synthesizing the equivalent virtual antenna array is shown in Fig. 3. The transmitter antennas transmit the chirp signal in turn and in cycle. Based on TDM-MIMO TX-RXs combination, a virtual array of 88 elements (rx1-rx88) was synthesized. Due to the number of virtual receiver antennas enhanced greatly, the angle resolution is improved in the developed radar system.

III. Improved High-Resolution Point Cloud Map Generation Scheme

In this section, the improved scheme for high-resolution PCM generation is described, which is accomplished by four steps.

A. Obtaining the Intermediate-Frequency Signals

First, the received signals from various targets are mixed with the transmitted signal to transform the wideband non-stationary signals into superposition of a train of single-tone signals, called intermediate-frequency (IF) signals. Followed by a lowpass filtering, i.e., the IF signals are expressed as:

$$x_{IF}(t) = \sum_{i=1}^{I} A_i \cos(2\pi S\tau_i t + 2\pi f_c \tau_i) + \omega(t) , \quad (4)$$

where I is the number of actual targets, A_i and τ_i are the echo signal amplitude and the TOF for the ith target, $\omega(t)$ is an additive white Gaussian noise signal. The IF signals then need to be sampled and transformed into the discretized signal sequences as follows:

$$x_{IF}(n) = \sum_{i=1}^{I} A_i \cos(2\pi(S\tau_i n T_s + f_c \tau_i)) + \omega(nT_s) , \quad (5)$$

where T_s and N denote the sampling interval and the number of samples, respectively, i.e., $N = T/T_s$, $0 \le n \le N\text{-}1$.

B. Estimating Range and Velocity Information of Targets

Due to the TOF of the ith target at the mth chirp signal $\tau_{i,m} = (2R_i + 2v_i mT)/c$, $0 \le m \le M\text{-}1$, i.e., (5) is re-written as:

$$x_{IF}(n,m) = \sum_{i=1}^{I} A_i \cos\ (4\pi(SR_i nT_s/c + f_c v_i mT/c + f_c R_i/c + Sv_i mTnT_s/c)) + \omega(n,m) , \quad (6)$$

where $x_{IF}(n, m)$ represents the discrete time samples of IF signals for the mth chirp signal, R_i and v_i are the range and radial velocity of the ith target, M is the numbers of chirp signals at one frame, and c is the light speed, respectively.

The range-frequency $f_r(i)$ and the Doppler-frequency $f_d(i)$ corresponding to the ith target are denoted as:

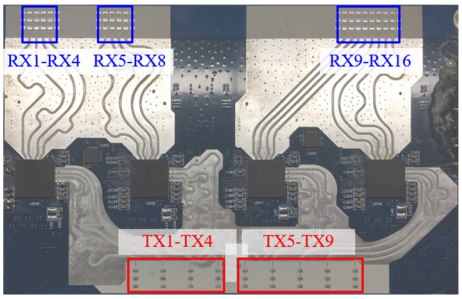

Fig. 2. Architecture of the MIMO antenna array of the radar system.

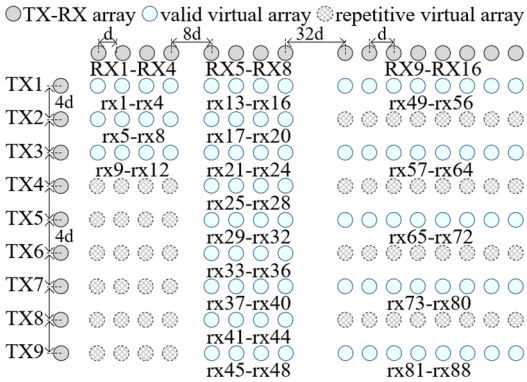

Fig. 3. Synthesis of the equivalent virtual array based on TDM-MIMO.

$$f_r(i) = 2SR_i/c,\ f_d(i) = 2f_c v_i/c\ . \quad (7)$$

From (7), the range information R_i and the velocity information v_i can be obtained using $f_r(i)$ and $f_d(i)$. Thus, two DFT operations are performed separately, as follows:

$$X[k_r, m] = \sum_{n=0}^{N\text{-}1} x_{IF}(n, m) exp(-j\frac{2\pi}{N} k_r n) , \quad (8)$$

$$Y[k_v, k_r] = \sum_{m=0}^{M\text{-}1} X[k_r, m] exp(-j\frac{2\pi}{M} k_v m) , \quad (9)$$

where k_r and k_v are the index of range-bins in range domain and Doppler-bins in Doppler domain, respectively. Equation (9) denotes a two-dimensional (2D) range-Doppler map (RDM) of single receiver antenna. The related 2D RDM of each receiver antenna will then be generated in turn.

C. Improved 2D CFAR Algorithm

To identify the effective targets and extract them from a clutter background, an improved 2D CFAR algorithm based on multi-channel fusion is proposed. Compared to the conventional cell-averaging (CA-CFAR) detector, the proposed algorithm is more effective in suppressing the noise floor and avoiding the masking effects in multi-target situations.

Firstly, all 2D RDMs from the 88 virtual receiver channels are fused, and can be expressed as:

$$Z(k_v, k_r) = \frac{1}{N_{rx}} \sum_{n_{rx}=0}^{N_{rx}\text{-}1} |Y[k_v, k_r, n_{rx}]|^2 , \quad (10)$$

where $Y[k_v, k_r, n_{rx}]$ indicates the 2D RDM from the n_{rx}th virtual receiver channel.

The improved 2D CFAR algorithm is then employed with a rectangular sliding reference window [6]. The local noise estimate in the immediate vicinity of the current cell under test (CUT) can be written as:

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$$\mathrm{U}(x, y) = \begin{cases} \frac{1}{L}\sum_{l=0}^{L-1} Z(l), & condA \\ \frac{1}{L}\left(\sum_{l=0}^{L-2} Z(l) + \mathrm{U}(x_1, y_1)\right), & condB \end{cases}, \quad (11)$$

where L is the number of reference cells in the 2D sliding reference window, (x, y) represents the coordinates of cell in RDM, *condA* represents a target exists at cell (x_1,y_1), and *condB* represents only noise background at cell (x_1,y_1).

It can then be determined whether the current CUT is a target based on the following conditions:

$$\begin{cases} Z(x, y) \le S(x, y), & H_0 \\ Z(x, y) > S(x, y), & H_1 \end{cases}, \quad (12)$$

with the detection threshold $S(x, y) = T \times \mathrm{U}(x, y)$, and T is a scaling factor. Where H_0 denotes only background noise, H_1 indicates that a target exists, $Z(x, y)$ is the power of the CUT. The proposed algorithm is summarized in Table I.

D. Estimating the AOA Based on Phase Difference

The sequence $\{Y(x_i, y_i, n_{rx}), n_{rx} = 0, 1, \ldots, N_{rx}\text{-}1\}$ can be extracted from 88 virtual receiver channels after target detection, which corresponds to the effective targets.

The phase of echo signal from the ith target at the n_{rx}th receiver antenna can be written as:

$$\phi_i(n_{rx}) = \phi_{i0} + 2\pi f_\theta(i) \cdot n_{rx}, \quad (13)$$

with the angle-frequency $f_\theta(i) = d\sin\theta_i/\lambda$, ϕ_{i0} is the initial phase. The third DFT is then conducted, as below:

$$P(i, k_\theta) = \sum_{n_{rx}=0}^{N_{rx}-1} Y(x_i, y_i, n_{rx}) exp(-j\frac{2\pi}{N_{rx}} k_\theta n_{rx}), \quad (14)$$

where $P(i,k_\theta)$ contains the angle information of target, k_θ is the index in angle domain. Through obtaining the angle information of targets, the high-resolution PCM can be drawn.

IV. Radar System Framework and Implementation

A block diagram and photographs of the newly developed 77 GHz FMCW radar system prototype are presented in Fig. 4(a) and 4(b), respectively. As shown in Fig. 4(a), the radar system is composed of an antenna array module, a transceiver front-end module, a data logging module, and a signal processing module. The antenna array is shown in Fig. 2. In the transceiver front-end module, the integrated transmitter chain consists of an oscillator of 40 MHz, a voltage-controlled oscillator, a closed-loop phase loop lock, and a frequency multiplier to generate the required FMCW signal, whose frequency ranges from 77 to 81 GHz coverage with 4 GHz bandwidth. The transmitted signal power is enhanced by the power amplifier. Each receiver chain consists of a low-noise amplifier, a mixer, a low-pass filter, and an analog digital converter to obtain the discrete IF signal sequences. The sampling data are captured in real time by the data logging module. Next, the logged data are transferred to a PC through Ethernet, and all algorithms are executed on the PC.

V. Field Test and Results Analysis

To verify the effectiveness of the proposed scheme for high-resolution PCM generation in multi-target and highly cluttered environments, the radar system was prepared for field test. The radar system consists of a 77 GHz FMCW radar transceiver module with antennas, a real-time data logging module, and a PC, as shown in Fig. 4(b). All procedures were implemented on PC using MATLAB. The test scenario is illustrated in Fig. 5. The detailed parameters of transmitted signals are presented in Table II.

TABLE I. The Improved 2D CFAR Algorithm

Algorithm: Target detection based on improved 2D CFAR
S1: All virtual receiver channels fusion: $Z(k_v, k_r)$;
S2: Estimate the local noise power $\mathrm{U}(x, y)$;
S3: Calculate the detection threshold $S(x, y) = T \times \mathrm{U}(x, y)$;
S4: Determine whether the current CUT is a target or not:
If $Z(x, y) > S(x, y)$ → A target exists, $Z(x, y) = \mathrm{U}(x, y)$
Else → No target
S5: All CUTs are detected, STOP; otherwise, go to **S2** to detect the next CUT.

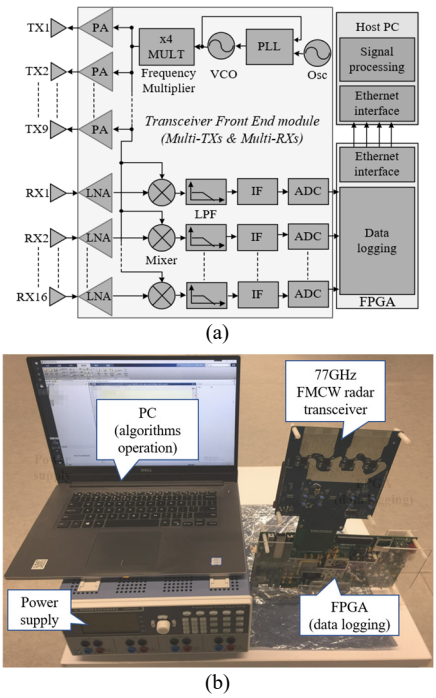

Fig. 4. 77 GHz FMCW radar system based on TDM-MIMO: (a) block diagram of the developed radar; (b) photograph of the developed radar.

In Fig. 5, the distance of the both two stationary reflectors (objects #1 and #2) from the radar is approximately $R_0 = 3.2$ m, and their distance interval is approximately $d_0 = 0.073$ m. Then, the AOA difference between them is $\theta_0 = 1.3022$ degrees. The distance of the third reflector (object #3) from the radar is approximately $R_1 = 6.2$ m and the corresponding AOA is $\theta_1 = 27.3$ degrees.

The measurement results for the test scenario are shown in Figs. 6-9. Fig. 6(a) shows the range-profile of one receiver channel after the range-DFT operation, where the x-axis represents the range (meters) and the y-axis represents the normalized magnitude (decibels). The distances of corresponding objects are estimated as $\hat{R}_0 = 3.188$ m, $\hat{R}_1 = 6.188$ m in Fig. 6(a), respectively, and the estimated error of distance is only 0.284%. Thus, the estimated distance closely coincides with the actual distance. Fig. 6(b) is the superposition of the range-profiles of 88 virtual receiver channels, which shows that the 88 virtual receiver channels are consistent in terms of their range-profiles.

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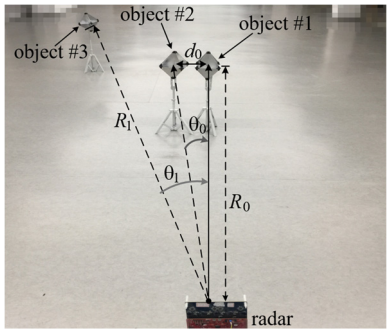

Fig. 5. Configuration of measurement scenario for the field test.

TABLE II. THE PARAMETERS OF TRANSMITTED SIGNALS

Parameter	Value
Starting frequency (GHz)	77
Transmitted bandwidth (GHz)	2.5
Frequency modulated slope (MHz/us)	50.018
Repetition period (us)	130
Number of chirps in one frame	9*256 = 2304
Sample size in one chirp	256

As can be seen in Fig. 7(a) and 7(b), three effective objects all appear in both figures. In Fig. 7(b), due to the use of the multi-channel fusion method, the noise floor is already reduced by approximately 5 dB in comparison to Fig. 7(a). The coordinates corresponding to effective targets become more obvious in Fig. 7(b) than in Fig. 7(a), which is more beneficial in accurately detecting effective targets in RDM.

Fig. 8(a) and 8(b) show the performance of target detection in a multi-target scenario, corresponding to a conventional CFAR detector and an improved CFAR detector, respectively. In Fig. 8(a), many ghost targets exist as well as the actual targets (#1, #2, and #3). However, in Fig. 8(b), three actual targets and only one ghost target exist. The accuracy rate, determining the number of effective targets, are 28.57% in Fig. 8(a) and 66.67% in Fig. 8(b). Thus, the proposed 2D CFAR detector improves the accuracy of target detection by approximately 1.33 times compared to the conventional CFAR detector.

Finally, the PCM for the test scenario was obtained, as shown in Fig. 9. Here, the x-axis is the cross range (meters) and the y-axis is the range (meters). Three effective objects (#1, #2, and #3) all appear in Fig. 9 as well as a ghost target. Moreover, the space coordinates of the effective objects in Fig. 9 are, respectively, (0m,3.188m); (-0.07246m, 3.188m); and (-2.832m, 5.501m). By calculation, the estimated angles are $\hat{\theta}_0 = 1.3021$ degrees and $\hat{\theta}_1 = 27.2401$ degrees. As $\theta_0 = 1.3022$ degrees in the test scenario, which also represents the theoretical angle resolution for the developed radar system, it is clear that $\hat{\theta}_0$ is approximately equal to θ_0. Thus, the conclusion can be drawn that, in the proposed scheme, the radar system can reach a theoretical angle resolution of 1.3022 degrees, and the estimated error of angle is only 0.114%. These results verify that the proposed scheme is effective to generate high-resolution PCMs.

VI. CONCLUSION

This paper proposes an improved scheme for high-resolution PCM generation based on the newly developed 77 GHz FMCW radar system prototype. The proposed scheme improves the angle resolution of the PCM by synthesizing an equivalent virtual antenna array and by applying an improved target detection algorithm. Compared with the conventional scheme, in multi-target environments, the proposed scheme significantly decreases the noise floor by approximately 5 dB in the range-Doppler map and improves the accuracy of target detection by 1.33 times. The angle resolution of the developed radar system has also reach to 1.3022 degrees.

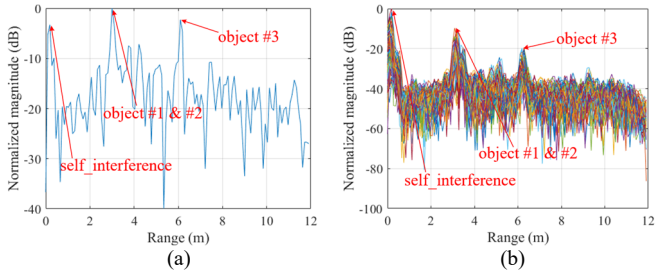

Fig. 6. Measurement results after the range-DFT operation: (a) one receiver channel; (b) superposition of the range-profiles of 88 virtual channels.

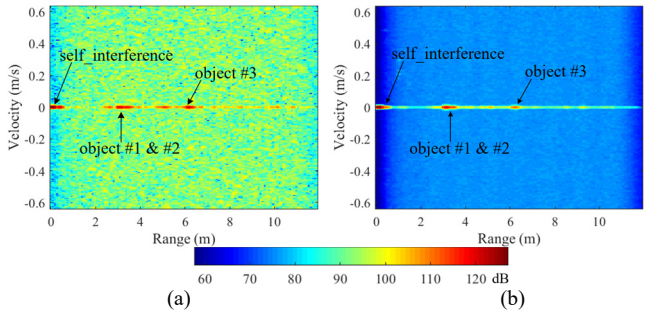

Fig. 7. Measurement results after the range-Doppler DFT operation: (a) one virtual receiver antenna; (b) multiple virtual receiver channels fusion.

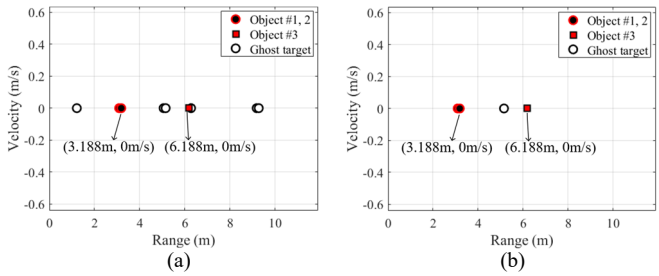

Fig. 8. Measurement results after executing the target detection algorithm: (a) conventional CA-CFAR detector; (b) improved 2D CFAR detector.

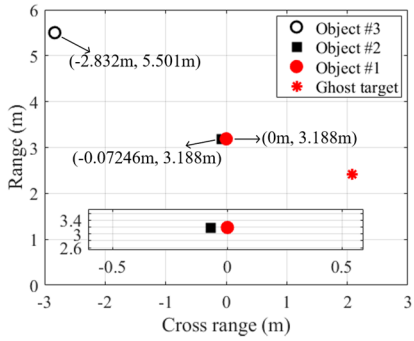

Fig. 9. The high-resolution point cloud map (PCM) for the test scenario.

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